Computer Organization (PCC-CS302)

Course Details

Credit Points: 3

Pre-Requisites

1. Concept of basic components of a digital computer, Basic concept of Fundamentals & Programme structures. Boolean Algebra

2. Basic number systems, Binary numbers, representation of signed and unsigned numbers, Binary Arithmetic as covered in Basic Computation & Principles of Computer Programming

3. Boolean Algebra

Syllabus Outline

Unit 1

- Basic organization of the stored program computer and operation sequence for execution of a program.
- Role of operating systems and compiler/assembler.
- Fetch, decode and execute cycle, Concept of operator, operand, registers and storage, Instruction format.
- Instruction sets and addressing modes.
- Commonly used number systems.
- Fixed- and floating-point representation of numbers.

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Unit 2

- Overflow and underflow.
- Design of adders - ripple carry and carry look ahead principles.
- Design of ALU.
- Fixed point multiplication - Booth's algorithm.
- Fixed point division - Restoring and non-restoring algorithms.
- Floating point - IEEE 754 standard.

Unit 3

- Memory unit design with special emphasis on implementation of CPU-memory interfacing.
- Memory organization, static and dynamic memory, memory hierarchy, associative memory.

- Cache memory, Virtual memory.
- Data path design for read/write access.

Unit 4

- Design of control unit - hardwired and microprogrammed control.
- Introduction to instruction pipelining.
- Introduction to RISC architectures.
- RISC vs CISC architectures.
- I/O operations - Concept of handshaking, Polled I/O, interrupt and DMA.

Computer Organization Lab (PCC-CS392)

Laboratory Experiments

1. Familiarity with IC-chips: a) Multiplexer, b) Decoder, c) Encoder, d) Comparator Truth Table verification and clarification from Data-book.
2. Design an Adder/Subtractor composite unit.
3. Design a BCD adder.
4. Design of a Carry-Look-Ahead Adder circuit.
5. Use a multiplexer unit to design a composite ALU.
6. Use ALU chip for multibit arithmetic operation,
7. Implement read write operation using RAM IC,
8. Cascade two RAM ICs for vertical and horizontal expansion.

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